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# Development of an Intelligent Assistant for Data Preparation Automation in Urban Transportation Management Systems

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Thesis for the Award of the Degree of  
MASTER  
7M06105 — Computer Science and Engineering

By  
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# Abstract

Urban transportation systems produce heterogeneous traffic and sensor data that must be cleaned, validated, and prepared before analysis. This thesis presents **Flowmatic**, an intelligent preparation and inference platform for batch uploads and smart-city pipelines, evaluated on a semi-synthetic Astana case study and public Hugging Face telemetry bundles.

Flowmatic implements six-dimensional data quality assessment, automated cleaning, and export in a NestJS/Vue Docker stack. A measured batch experiment ingests 30 000 Astana records in 406 ms with quality score 100/100. A smart-city workbench connects an Astana geospatial simulator, Manual/Auto model routing over a documented registry, and medallion lake export.

Eight neural checkpoints—PatchTST, iTransformer, TranAD, TimesBlock, STGCN, SAITS, a Transformer severity classifier, and DLinear—are trained under a shared temporal hold-out protocol (70/15/15 split, three seeds). The severity classifier reaches  $\text{macro-F1} = 0.911 \pm 0.017$  on the test partition; TranAD reconstruction RMSE is  $0.035 \pm 0.004$  on Astana windows. Weights and reproduction scripts are published on Hugging Face ([@pushthetempo/flowmatic-\\*](https://huggingface.co/pushthetempo/flowmatic-*/)).

The dissertation demonstrates that preparation, multimodal inference, and operational deployment can be delivered as one auditable platform with reproducible artifacts.

**Keywords:** Data Preparation, Intelligent Transportation Systems, Smart Cities, Model Routing, Time Series, Anomaly Detection, Flowmatic

# Dedication and acknowledgements

I dedicate this work to my family and to the faculty and colleagues at Astana IT University who supported this research.

I acknowledge the use of open traffic benchmarks, Hugging Face model hosting, and the Flowmatic open-source platform developed during this thesis.

This research was conducted as part of the project “Intelligent models and methods of Smart City digital ecosystem for sustainable development and the citizens’ quality of life improvement,” funded by the Committee of Science of the Ministry of Science and Higher Education of the Republic of Kazakhstan (Grant No. BR24992852).

# Student's declaration

I, Ramazan Bakytuly certify that the work embodied in this research/project work, carried out by me under the supervision of Aivar Sakhipov Aituarovich for the period of September 2023 to June 2026 at Astana IT University. The work has not been submitted for the award of any other degree/ diploma except where due acknowledgement has been made in the text.

I further declare that the material obtained from other sources has been duly acknowledged in the thesis.

SIGNED: ..... DATE: .....

# Certificate by Supervisor

This is to certify that research work embodied in this thesis entitled “Development of an Intelligent Assistant for Data Preparation Automation in Urban Transportation Management Systems” submitted to Astana IT University, for the award of the degree of Master (7M06105 Computer Science and Engineering) has been carried out by Ramazan Bakytuly under my supervision at Astana IT University, Astana from September 2023 to June 2026.

To the best of my knowledge and belief, this work is original and has not been submitted so far in part or in full for the award of any degree or diploma of any University/ Institute.

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# Chapter 1

## Introduction

### 1.1 Relevance, Problem Statement, and Research Gap

Urban transportation management systems rely on heterogeneous data collected from sensors, CSV exports, API feeds, and streaming telemetry [1–3]. In practice, these sources often contain missing values, duplicated records, schema inconsistencies, invalid coordinates, delayed timestamps, and domain-specific anomalies. Such defects complicate forecasting, anomaly detection, classification, and other analytical workflows, because the reliability of downstream results depends directly on the quality and reproducibility of the preparation process [? ].

Existing studies on automated preprocessing, intelligent transportation systems, and neural traffic analytics usually address these stages separately [4–8]. Data cleaning is often treated as a preliminary technical step, while model training, deployment, and monitoring are evaluated independently [? ? ]. As a result, transportation analysts and system operators may lack a unified mechanism for measuring data quality, applying consistent transformations, preserving preparation history, and connecting prepared datasets to downstream models.

The research problem addressed in this thesis is the lack of an auditable and reproducible workflow for automating data preparation in urban transportation management systems while preserving compatibility with downstream analytical models. This thesis investigates how such a workflow can be designed, implemented, and evaluated through

the **Flowmatic** platform.

## 1.2 Aim and Objectives of the Research

The aim of this research is to develop and evaluate an intelligent software platform for automating data preparation in urban transportation management systems, with support for measurable data quality assessment, reproducible preprocessing, and integration with downstream analytical models.

To achieve this aim, the following objectives are defined:

1. To analyze existing approaches to data quality assessment, automated preprocessing, and model lifecycle management in intelligent transportation systems.
2. To design the architecture of a platform that supports batch and streaming data preparation workflows for urban transportation data.
3. To implement mechanisms for data quality assessment, cleaning, transformation, storage, and export.
4. To integrate the preparation workflow with a model registry and inference service for transportation-related analytical tasks.
5. To evaluate the proposed platform using data quality metrics, latency measurements, controlled corruption experiments, baseline comparisons, and routing validation.
6. To define the limitations of the proposed approach and identify directions for future deployment in real urban transportation environments.

The object of the research is the data preparation process in urban transportation management systems. The subject of the research is the set of methods and software mechanisms for automated quality assessment, preprocessing, routing, and preparation of transportation data for analytical models.

### 1.3 Scientific Novelty and Practical Significance

The scientific novelty of the research lies in the integrated evaluation of data preparation, quality assessment, and model-oriented processing within a single reproducible workflow for urban transportation data.

The main elements of novelty are:

1. An integrated preparation-to-inference lifecycle that combines data quality assessment, automated cleaning, export, model registry usage, and downstream inference in one workflow.
2. A six-dimensional Data Quality Index applied to urban transportation data and evaluated under controlled data corruption scenarios.
3. A reproducible evaluation protocol linking preparation quality, batch-processing latency, baseline comparisons, neural model performance, and routing validation.
4. A model-oriented routing mechanism that connects prepared traffic and sensor streams to task-specific analytical models.
5. Published artifacts and evaluation outputs that support reproducibility of the experimental results [? ? ].

The practical significance of the research is represented by the implemented Flowmatic platform, which provides authenticated data upload, automated quality assessment, cleaning, storage, export, smart-city simulation workflows, model registry integration, and inference support. The platform can be used as a research prototype for preparing transportation datasets and demonstrating how data preparation can be connected with downstream analytics in urban transportation management systems.

### 1.4 Thesis Structure

The thesis is organized as follows. Chapter 2 reviews related work on data quality, automated preprocessing, intelligent transportation systems, forecasting, anomaly detection, model registries, and reproducibility. Chapter 3 describes the research methodology, including the data quality assessment approach, preparation workflow, routing logic,

evaluation protocols, and research questions. Chapter 4 presents the implementation of the Flowmatic platform. Chapter 5 reports experimental results and discusses preparation quality, latency, baseline comparisons, neural evaluation, and routing validation. Chapter 6 summarizes the findings, limitations, and future research directions. Appendix B documents reproducibility commands.

# Chapter 2

## Literature Review

### 2.1 Scope of the Review

This chapter reviews work relevant to **Flowmatic**: assessing traffic data quality, automating preparation, deploying stream-oriented ITS platforms, and evaluating neural models for forecasting, reconstruction, imputation, and classification.

### 2.2 Data Quality in Transportation Analytics

General data quality research treats quality as fitness for use [1, 2]. Transportation streams are spatially distributed, time ordered, and heterogeneous [9, 10]. Composite indices [11] aggregate dimensions when weights remain application dependent. Flowmatic implements six dimensions—completeness, consistency, accuracy, timeliness, uniqueness, and validity—in `QualityService` (Table 3.1).

### 2.3 Automated Preprocessing and Missing Data

Automated preprocessing adapts operations to data and downstream tasks [4, 5, 12]. Missing traffic data requires temporal and spatial structure [13, 14]. Phase 2 bundles in this thesis materialise windows for neural training; SAITS handles masked reconstruction within the production portfolio.

## 2.4 Feature Engineering and Representation Learning

Traffic analytics benefit from cyclical time features, rolling statistics, and graph structure [15–17]. Flowmatic generates windowed tensors and adjacency for STGCN while keeping schema-level features auditable in batch uploads.

## 2.5 Anomaly Detection and Classification

Classical outlier methods (LOF, Isolation Forest) and deep reconstruction models address different anomaly types [18–21]. Flowmatic deploys **TranAD-style** reconstruction on Astana streams (Equation 3.6) and a **Transformer** severity classifier with macro-F1 reporting (Equation 3.8).

## 2.6 Intelligent Transportation Systems and Deployment

Smart-city architectures combine sensing, streaming, and analytics [3, 6, 22]. Flowmatic uses HTTP/WebSocket ingestion, queue workers, and a model-inference microservice rather than a notebook-only workflow.

## 2.7 Explainability

Agencies require traceable diagnostics [23, 24]. Flowmatic emphasises deterministic Insight Engine metrics plus neural residuals; optional LLM narration is constrained to pipeline context.

## 2.8 Neural Time-Series Models

Decomposition linear models [25], PatchTST [7], and STGCN [8] motivate the eight-checkpoint portfolio evaluated under a single temporal split protocol (Chapter 5).

## 2.9 Scope of This Thesis

The dissertation reports completed work only:

1. **Flowmatic platform** — batch preparation, smart-city workbench, registry routing, Docker deployment;
2. **Neural portfolio** — eight Hugging Face checkpoints with multi-seed held-out test metrics and streaming latency benchmarks.

All quantitative claims in Chapters 5–6 refer to artifacts in this repository or published Hub weights.

# Chapter 3

## Methodology

### 3.1 Overview

The methodology covers the **Flowmatic** preparation pipeline, the **production model registry**, and the **neural evaluation protocol** used in Chapter 5. Batch upload and smart-city deployment modes are illustrated in Chapter 4 (Figures 4.1 and 4.2).

### 3.2 Research Questions

The evaluation programme in Chapter 5 addresses the following research questions:

**RQ1:** Does the Flowmatic preparation workflow deliver measurable quality outcomes (completeness, consistency, latency) on urban traffic uploads before downstream inference?

**RQ2:** Can the preparation workflow be operationalised as a reproducible service with measured end-to-end latency and quality reporting?

**RQ3:** How do the eight production neural models perform across forecasting, reconstruction, imputation, and classification under a common temporal hold-out protocol?

**RQ4:** Does a registry-based Manual/Auto routing policy allow modality-appropriate model selection in a smart-city pipeline?

### 3.3 Evaluation Scope and Validity

Two validation layers must not be conflated. *Batch platform validation* is demonstrated on the Astana CSV export through the Nest ingestion API (latency, quality score, cleaning). *Neural training validation* uses Astana plus Hugging Face weather, ETT, and traffic-graph bundles prepared offline; those bundles are not all re-uploaded through the Nest API in this thesis. METR-LA and PEMS datasets are excluded from the evaluated dataset table because hold-out experiments were not completed for them in this study.

Experiments use semi-synthetic Astana data with rule-based severity labels. The smart-city demo uses simulated feeds rather than a production agency deployment. External municipal incident-label AUROC, expert-labelled routing accuracy, and live agency deployment remain future work; detailed limitations are discussed in Chapter 6.

### 3.4 Data Quality Assessment

Flowmatic computes a weighted Data Quality Index (DQI) for each upload and pipeline run:

$$\text{DQI} = \frac{\sum_{i=1}^6 w_i Q_i}{\sum_{i=1}^6 w_i}, \quad Q_i \in [0, 1], \quad (3.1)$$

over completeness, consistency, accuracy, timeliness, uniqueness, and validity. Completeness follows  $C = 1 - |M|/|D|$ ; timeliness uses exponential decay  $T = \exp(-\lambda \bar{a})$ . Table 3.1 lists default weights  $w_i = 1/6$ . The NestJS `QualityService` implements these dimensions via schema typing, duplicate detection, valid-range checks, and Z-score outlier flags.

Table 3.1: DQI dimension weights and score definitions (Flowmatic `QualityService`).

| Dim. | Criterion    | Weight $w_i$ | Score $Q_i$                             |
|------|--------------|--------------|-----------------------------------------|
| 1    | Completeness | 1/6          | $1 -  M / D $                           |
| 2    | Consistency  | 1/6          | Rule-pass rate on schema checks         |
| 3    | Accuracy     | 1/6          | Valid-range compliance                  |
| 4    | Timeliness   | 1/6          | $\exp(-\lambda \bar{a})$ on arrival lag |
| 5    | Uniqueness   | 1/6          | 1 – duplicate-row fraction              |
| 6    | Validity     | 1/6          | Type and constraint satisfaction        |

## 3.5 Preparation and Feature Windows

After quality assessment, workers apply imputation for missing numeric fields, duplicate removal, and type coercion. For neural training and inference, each sensor stream is windowed into tensors  $\mathbf{X} \in \mathbb{R}^{L \times d}$  with  $L = 48$  by default. Cyclical time encodings

$$f_{\sin}(t) = \sin(2\pi t/P), \quad f_{\cos}(t) = \cos(2\pi t/P) \quad (3.2)$$

and traffic-derived channels (density, speed, severity labels) are materialised in Phase 2 dataset bundles (`run_phase2_prep.py`). Graph models additionally receive adjacency  $\mathbf{A}$  for STGCN.

## 3.6 Adaptive Core Unit and Model Registry

### 3.6.1 Registry

Each production checkpoint  $m$  exposes metadata: kind (e.g. `tranad_anomaly`, `transformer_classifier`), dataset affinity, tasks, sensor kinds, optional geo requirement, and priority. The registry loads from local `metadata.json` and the Hugging Face manifest (Table 4.2).

### 3.6.2 Routing policy

For sensor kind  $k$  and payload features  $\mathbf{f}$ , the router builds an event profile (modality, preferred task, geo flag). In **Manual** mode the operator selects model  $m^*$ . In **Auto** mode:

$$m^* = \arg \min_{m \in \mathcal{M}_k} \pi(m \mid \mathbf{f}), \quad (3.3)$$

where  $\mathcal{M}_k$  filters incompatible modalities/tasks and  $\pi$  is a priority score (lower is better). Anomaly-capable tasks receive precedence when anomaly detection is enabled.

---

**Algorithm 1** Rule-based model routing (Auto mode)

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- 1: **Input:** sensor kind  $k$ , payload  $\mathbf{f}$ , registry  $\mathcal{M}$ , anomaly flag  $a$
  - 2:  $profile \leftarrow \text{CLASSIFY}(k, \mathbf{f})$
  - 3: **if**  $a$  **and**  $\exists m \in \mathcal{M}$  with task anomaly matching  $profile$  **then**
  - 4:     **return** best-priority anomaly model (TranAD slot)
  - 5: **end if**
  - 6:  $\mathcal{C} \leftarrow \{m \in \mathcal{M} : \text{modality}(m) = \text{profile.modality} \wedge \text{task}(m) \supseteq \text{profile.task}\}$
  - 7: **return**  $\arg \min_{m \in \mathcal{C}} \text{priority}(m)$
-

### 3.7 Neural Task Formulations

Each production model consumes a normalized window  $\mathbf{X} \in \mathbb{R}^{L \times d}$  of length  $L = 48$  and feature dimension  $d$ . Targets are denoted  $y$ ; graph models use adjacency  $\mathbf{A} \in \mathbb{R}^{N \times N}$  over  $N$  sensors.

#### 3.7.1 Forecasting (PatchTST, iTransformer, DLinear, TimesBlock, STGCN)

Point forecasting minimizes squared error on a held-out temporal test split:

$$\mathcal{L}_{\text{forecast}} = \frac{1}{|\mathcal{T}_{\text{test}}|} \sum_{t \in \mathcal{T}_{\text{test}}} \|\hat{y}_t - y_t\|_2^2. \quad (3.4)$$

DLinear decomposes each channel into trend and seasonal components via moving-average smoothing kernel  $\mathbf{1}_k/k$  before applying linear projections [25]. PatchTST partitions  $\mathbf{X}$  into patches of length  $p$  and applies channel-independent self-attention over patch tokens [7]. STGCN combines graph convolution on  $\mathbf{A}$  with temporal convolution:

$$\mathbf{H}^{(l+1)} = \sigma(\mathbf{A}\mathbf{H}^{(l)}\mathbf{W}_g^{(l)}) * \mathbf{W}_t^{(l)}, \quad (3.5)$$

where  $*$  denotes temporal convolution and  $\sigma$  is ReLU.

#### 3.7.2 Anomaly detection (TranAD-style reconstruction)

Let  $f_\theta$  reconstruct window  $\mathbf{X}$ . Anomaly score is reconstruction residual:

$$s(\mathbf{X}) = \frac{1}{Ld} \sum_{i=1}^L \sum_{j=1}^d \left( X_{ij} - \hat{X}_{ij} \right)^2. \quad (3.6)$$

Training minimizes  $\|\mathbf{X} - f_\theta(\mathbf{X})\|_F^2$  on normal traffic windows; elevated  $s(\mathbf{X})$  flags corrupted streams.

#### 3.7.3 Imputation (SAITS-style masked reconstruction)

Given mask  $\mathbf{M} \in \{0, 1\}^{L \times d}$  with mask ratio  $\rho = 0.25$ , the model predicts occluded values:

$$\mathcal{L}_{\text{impute}} = \frac{\sum_{i,j} (1 - M_{ij})(X_{ij} - \hat{X}_{ij})^2}{\sum_{i,j} (1 - M_{ij})}. \quad (3.7)$$

### 3.7.4 Severity classification (Transformer encoder)

For class set  $\mathcal{C}$ , the encoder outputs logits  $\mathbf{z} = h_\phi(\mathbf{X})$  and probabilities  $\pi_c = \text{softmax}(\mathbf{z})_c$ .

We report macro-F1:

$$F_1^{\text{macro}} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \frac{2P_c R_c}{P_c + R_c}. \quad (3.8)$$

## 3.8 Neural Evaluation Methodology

All eight checkpoints share temporal splitting (Table 5.3 in Chapter 5), z-score standardisation fit on training windows only, and seed-controlled training (`phase3_multiseed_suite.yaml`). Objectives are defined in Section 3.7. Reported metrics use only the held-out 15% test partition; multi-seed aggregation is  $\bar{m} \pm 1.96 s / \sqrt{n}$  with  $n = 3$ .

## 3.9 Insight Engine and Copilot

The Insight Engine composes deterministic analytics (hotspots, drift, correlation) with optional LLM narration constrained to retrieved pipeline context. Neural models provide primary diagnostics: reconstruction residuals (TranAD, Equation 3.6), imputation masks (SAITS), and classification logits (Transformer severity).

# Chapter 4

## Implementation

### 4.1 System Architecture

Flowmatic is a TypeScript monorepo: NestJS backend (Prisma/PostgreSQL), Vue 3 frontend, Bun/Python inference sidecar, and Docker Compose. Figure 4.3 shows PostgreSQL, MinIO, RabbitMQ, `sensor-simulator`, `model-inference`, Nest backend, and Nginx-served SPA.

### 4.2 Batch Preparation Module

Clients upload CSV/JSON via `IngestionController`. Files land in S3-compatible storage; workers parse, score quality, clean, and persist summaries. Export adapters target PostgreSQL, MongoDB, Hugging Face datasets, and flat files. Figure 4.1 corresponds to Table 4.1.

### 4.3 Smart City Module

Organisation-scoped pipelines use four stages:

1. **Sources** — HTTP/WebSocket simulators or connectors;
2. **Core unit** — validation, Manual/Auto routing, inference;
3. **Lake & export** — bronze/silver/gold prefixes on MinIO (Figure 4.4);

#### 4. Federated demo — optional coordinator metadata.

WebSockets stream events to the UI; queue workers handle asynchronous export. Event-driven semantics match ITS practice without mandating a particular message broker.

### 4.4 Model Inference Service

`model-inference` loads TorchScript checkpoints from `models/` or Hugging Face Hub (Table 4.2). Requests specify checkpoint id, feature window  $L=48$ , and task tensors (e.g. **A** for STGCN). Latency benchmarks appear in Section 5.6.

### 4.5 Research Artifact Pipeline

Training lives under `models/`: `run_phase2_prep.py` builds datasets; `run_phase3_core.py` runs the multi-seed suite, streaming benchmarks, and Hub upload. Each checkpoint ships `metadata.json`, scaler statistics, metrics JSON, and TorchScript exports.

### 4.6 User Interface

Figures 4.6–4.8 show the Astana Geospatial Demo: sensor sources, routing mode, pipeline rail, and workflow graph.

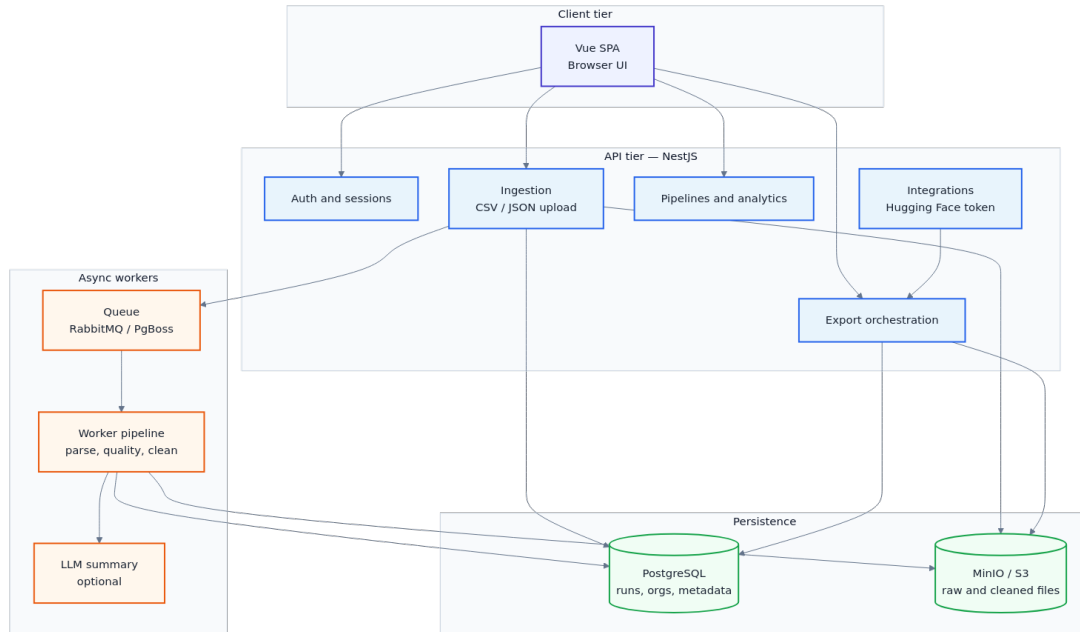


Figure 4.1: Batch upload pipeline (NestJS + Vue).

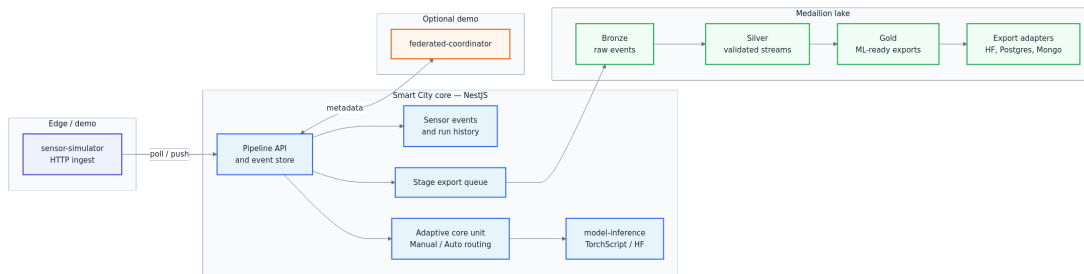


Figure 4.2: Smart-city pipeline with adaptive core unit and medallion lake.

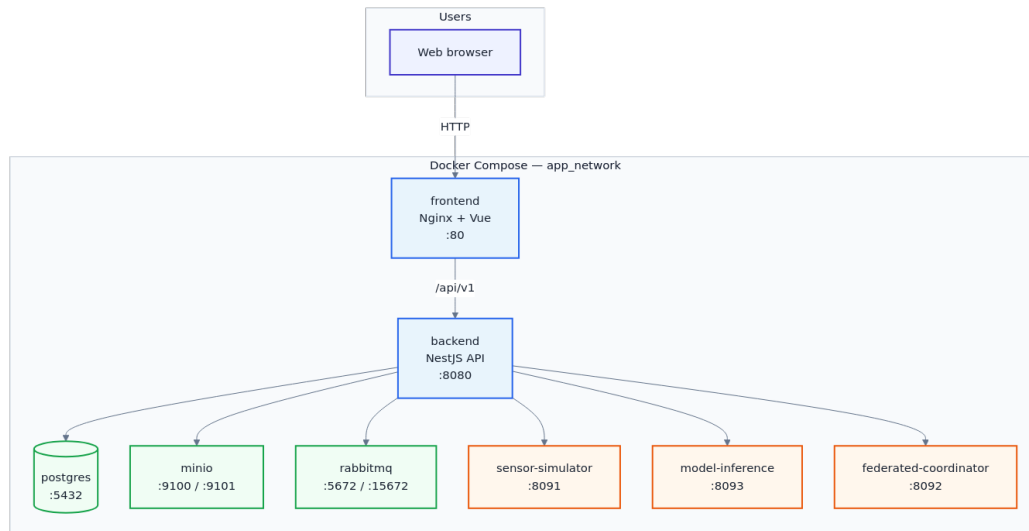


Figure 4.3: Docker Compose deployment topology.

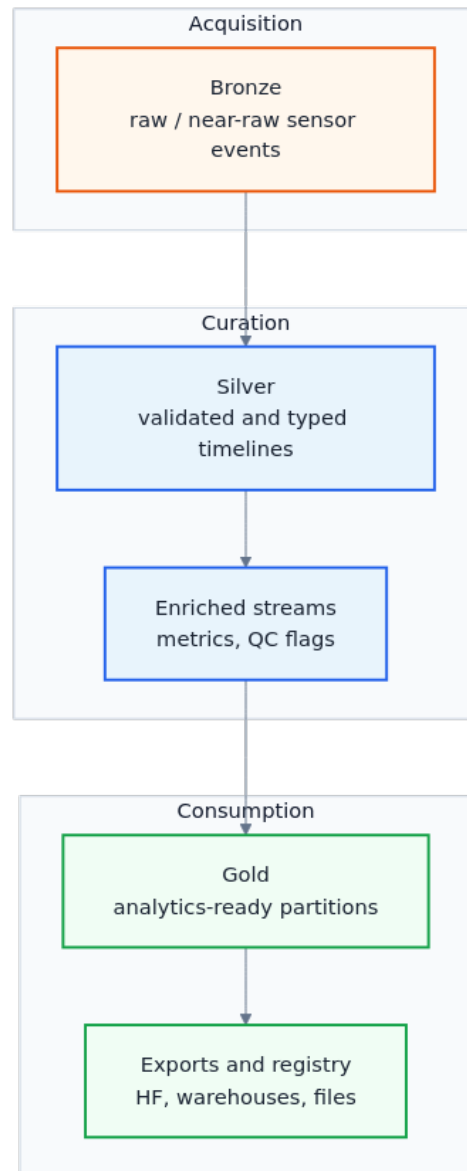


Figure 4.4: Medallion data lake layout (bronze, silver, gold).

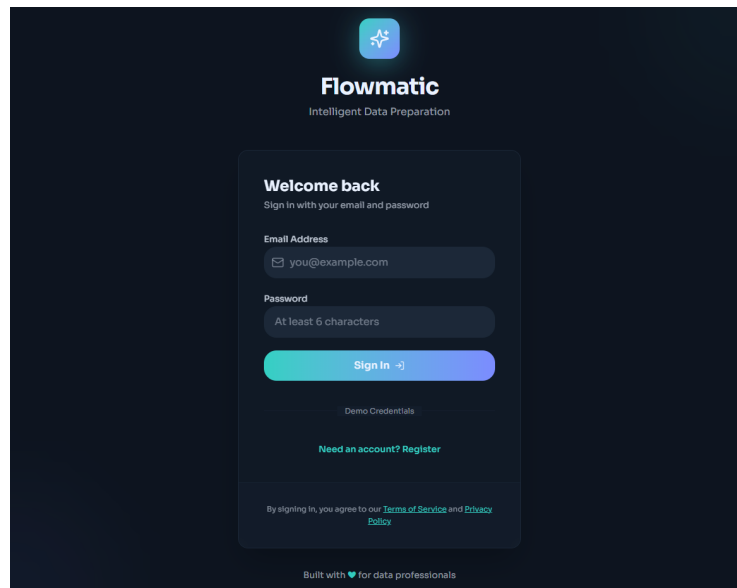


Figure 4.5: Flowmatic login interface.

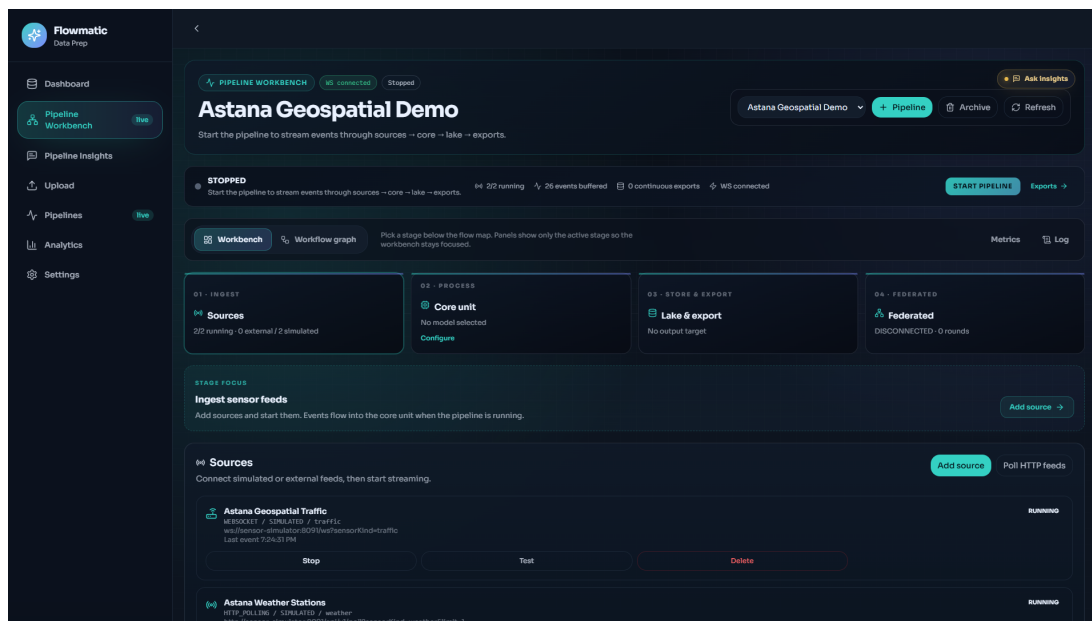


Figure 4.6: Astana Geospatial Demo workbench with simulated traffic and weather sources.

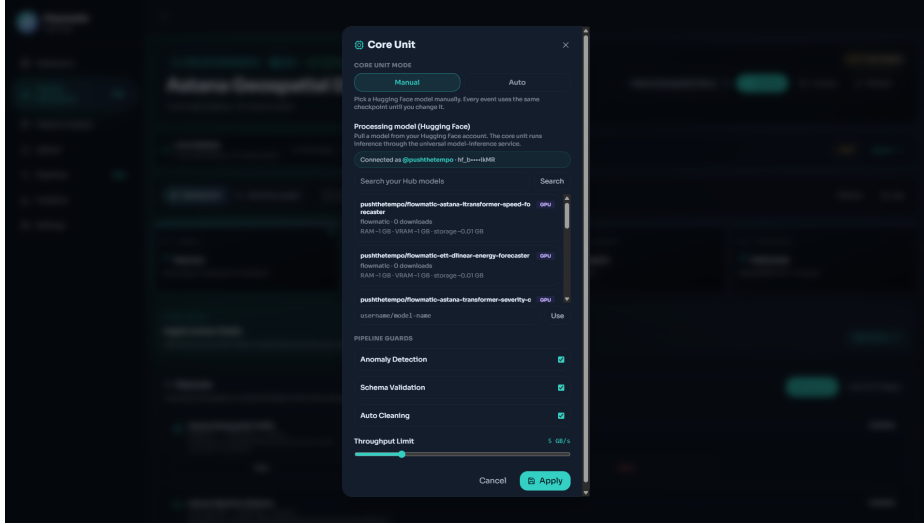


Figure 4.7: Adaptive core unit configuration with Manual/Auto routing and Hugging Face model catalogue.

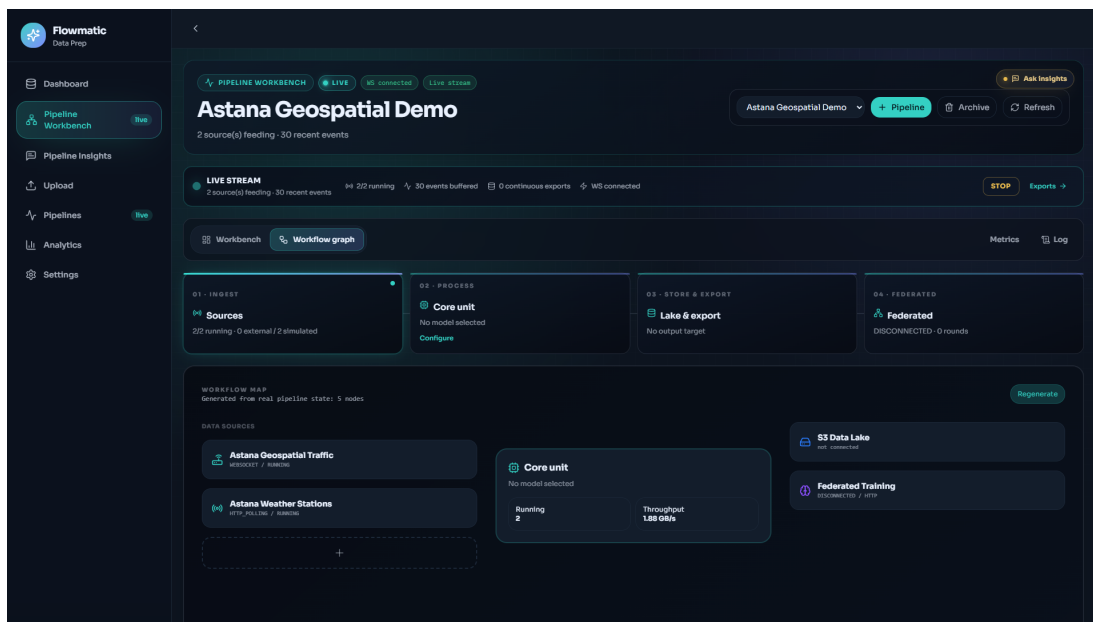


Figure 4.8: Workflow graph for the Astana geospatial simulator pipeline.

## 4.7 Implementation Summary

Flowmatic unifies preparation workers, the eight-model registry, and inference behind authenticated APIs. Table 4.2 maps each deployed slot to its Hugging Face repository; Table 4.1 anchors the batch path to a measured production run.

Table 4.1: End-to-end batch upload experiment on `astana_synthetic_data.csv` via Flowmatic production API (May 2026).

| Metric                                      | Value                            |
|---------------------------------------------|----------------------------------|
| Records ingested / cleaned                  | 30 000 / 30 000                  |
| Source size                                 | 2.21 MB                          |
| Processing time                             | 406 ms                           |
| Quality score (initial $\rightarrow$ final) | 100 / 100                        |
| Schema insight                              | 5 numeric, 4 categorical columns |
| Anomalies detected                          | None (proceed to analytics)      |
| Total wall-clock (auth + upload + poll)     | 2.3 s                            |

Table 4.2: Production neural portfolio registered in Flowmatic and published on Hugging Face (May 2026).

| Task slot                | Architecture | Dataset    | HF repository                                               |
|--------------------------|--------------|------------|-------------------------------------------------------------|
| Traffic density forecast | PatchTST     | Astana     | <code>flowmatic-astana-patchtst-density-forecaster</code>   |
| Speed forecast           | iTransformer | Astana     | <code>flowmatic-astana-itransformer-speed-forecaster</code> |
| Anomaly reconstruction   | TranAD       | Astana     | <code>flowmatic-astana-tranad-anomaly-detector</code>       |
| Weather forecast         | TimesBlock   | HF weather | <code>flowmatic-weather-timesblock-forecaster</code>        |
| Graph traffic forecast   | STGCN        | HF traffic | <code>flowmatic-traffic-stgcn-forecaster</code>             |
| Imputation               | SAITS        | Astana     | <code>flowmatic-astana-saits-imputer</code>                 |
| Severity classification  | Transformer  | Astana     | <code>flowmatic-astana-transformer-severity-classif</code>  |
| Energy forecast          | DLinear      | HF ETT     | <code>flowmatic-ett-dlinear-energy-forecaster</code>        |

Namespace prefix: `pushthetempo/`. Served by `model-inference` via TorchScript or Hub download.

Table 4.3: Smart-city live demo pipeline (*Astana Live Demo*) with simulated sensor sources.

| Parameter                         | Observed value                      |
|-----------------------------------|-------------------------------------|
| Running sources                   | 2 / 2 (IoT + weather, HTTP polling) |
| Simulator endpoint                | <code>sensor-simulator:8091</code>  |
| WebSocket status                  | Connected                           |
| Recent events (UI buffer)         | 25+                                 |
| Poll interval (upload experiment) | 1.9 s                               |
| Federated coordinator             | Disconnected (demo optional)        |

# Chapter 5

## Results and Discussion

### 5.1 Overview

This chapter answers the research questions in Section 3.2 through four evaluation layers:

1. **Preparation and quality (RQ1)** — measured outcomes on batch upload (DQI, cleaning, latency).
2. **Platform validation (RQ2)** — reproducible API experiments and the live Astana smart-city workbench.
3. **Neural portfolio (RQ3)** — eight Hugging Face checkpoints on held-out temporal test splits with three seeds ( $s \in \{42, 7, 2026\}$ ), ablations, transfer, and streaming latency.
4. **Routing (RQ4)** — Manual/Auto registry behaviour in the workbench (quantitative router ablations deferred).

Neural results are reproduced with `python models/paper/run_phase3_core.py`. Checkpoints are listed in Table 4.2.

Table 5.1: Contribution matrix: claim, evidence artifact, and research question.

| Contribution        | Evidence                 | RQ                    | Limit                    |
|---------------------|--------------------------|-----------------------|--------------------------|
| DQI + batch prep    | Table 4.1; RQ1, RQ2      | Semi-synthetic Astana |                          |
| Smart-city pipeline | Table 4.3; Figs. 4.5–4.8 | RQ2, RQ4              | Simulated sensors only   |
| Neural portfolio    | Table 5.4; HF manifest   | RQ3                   | METR-LA hold-out pending |
| Model registry      | Table 4.2; Algorithm 1   | RQ3, RQ4              | Router metrics pending   |
| Reproducibility     | Appendix B               | All                   | Three seeds ( $n=3$ )    |

## 5.2 Experimental Protocol

### 5.2.1 Datasets

Table 5.2 lists dataset families for neural training. Astana is the primary smart-city case study. Hugging Face bundles supply weather, energy, and traffic-graph telemetry. METR-LA and PEMS-BAY are ingested for planned external validation.

Table 5.2: Experimental datasets used in Flowmatic Phase 2–3 evaluation (May 2026).

| Key          | Role                    | Rows used     | Features | Source                        |
|--------------|-------------------------|---------------|----------|-------------------------------|
| astana       | Primary smart-city case | 18 000–30 000 | 9        | Semi-synthetic Astana traffic |
| hf_weather   | Weather forecast        | 18 000        | 22       | Hugging Face weather bundle   |
| hf_ett       | Energy / utility proxy  | 16 000        | 8        | ETTh1 (HF)                    |
| hf_traffic   | Graph traffic           | 8 000         | 863      | HF traffic sensors            |
| pems_metr_la | External traffic graph  | 12 000        | 207      | METR-LA (HF parquet)          |
| pems_bay     | External traffic graph  | 12 000        | 325      | PEMS-BAY (HF parquet)         |

Table 5.3: Temporal data split protocol (identical across neural runs).

| Split      | Share          | Use                                                   |
|------------|----------------|-------------------------------------------------------|
| Train      | 70% (earliest) | Parameter estimation                                  |
| Validation | 15%            | Early stopping / model selection                      |
| Test       | 15% (latest)   | <b>Reported metrics only</b> — never used in training |

Table 5.4: Production portfolio: multi-seed test metrics (seeds 42, 7, 2026; mean  $\pm$  95% CI).

| Task slot        | Model        | Metric     | Mean  | 95% CI      |
|------------------|--------------|------------|-------|-------------|
| Traffic forecast | PatchTST     | RMSE       | 0.561 | $\pm 0.004$ |
| Traffic anomaly  | TranAD       | RMSE       | 0.035 | $\pm 0.004$ |
| Weather forecast | TimesBlock   | RMSE       | 0.034 | $\pm 0.002$ |
| Graph traffic    | STGCN        | RMSE       | 0.479 | $\pm 0.002$ |
| Imputation       | SAITS        | Masked MSE | 0.640 | $\pm 0.009$ |
| Severity class.  | Transformer  | Macro-F1   | 0.911 | $\pm 0.017$ |
| Energy forecast  | DLinear      | RMSE       | 0.153 | $\pm 0.011$ |
| Speed forecast   | iTransformer | RMSE       | 0.999 | $\pm 0.001$ |

Table 5.5: Sequence-length ablation (Astana PatchTST, ETT DLinear; test RMSE).

| Model    | Dataset         | $L = 24$ | $L = 96$ |
|----------|-----------------|----------|----------|
| PatchTST | Astana density  | 0.559    | 0.807    |
| DLinear  | ETT energy (OT) | 0.241    | 0.147    |

Table 5.6: Cross-dataset transfer (train on source, evaluate on target test split; RMSE).

| Architecture | Train      | Test       | RMSE  |
|--------------|------------|------------|-------|
| DLinear      | hf_ett     | hf_weather | 0.165 |
| DLinear      | hf_weather | hf_ett     | 0.634 |
| PatchTST     | hf_ett     | hf_weather | 0.291 |
| PatchTST     | hf_weather | hf_ett     | 0.492 |

### 5.2.2 Hold-out validation

Neural models never observe the test partition during training. Windows follow temporal order: 70% train, 15% validation, 15% test (Table 5.3). RMSE, masked MSE, and macro-F1 are computed on the test split only.

### 5.2.3 Multi-seed robustness

Each architecture is trained for seeds 42, 7, and 2026 with shared hyper-parameters (`phase3_multiseed_suite.yaml`). We report  $\bar{m} \pm 1.96 s / \sqrt{n}$  ( $n = 3$ ).

## 5.3 Preparation and Quality Outcomes

Table 4.1 reports an end-to-end batch run on `astana_synthetic_data.csv`: 30 000 records ingested and cleaned in 406 ms, quality score 100/100, and schema profiling (five numeric and four categorical columns). No blocking anomalies were flagged on this upload. These outcomes answer **RQ1** and **RQ2** for the batch path: preparation is automated, scored, and fast enough for interactive analyst use. Figure 4.1 matches the measured Docker Compose deployment.

## 5.4 Platform Case Studies

### 5.4.1 Live smart-city workbench

Table 4.3 summarises the Astana Live Demo: two HTTP sensor sources (traffic and weather), WebSocket event delivery, four pipeline stages, and Manual/Auto routing on the core unit. Figures 4.6–4.8 in Chapter 4 show the captured interface. This extends **RQ2** from batch uploads to streaming operation.

## 5.5 Neural Production Portfolio

### 5.5.1 Portfolio

Eight checkpoints in Table 4.2 cover forecasting (PatchTST, iTransformer, TimesBlock, STGCN, DLinear), anomaly reconstruction (TranAD), imputation (SAITS), and severity classification (Transformer). Formulations appear in Section 3.7.

### 5.5.2 Multi-seed test performance

Table 5.4 aggregates held-out test metrics:

- **Severity classification (Transformer):**  $\text{macro-F1} = 0.911 \pm 0.017$ .
- **Anomaly reconstruction (TranAD):** test RMSE =  $0.035 \pm 0.004$  on normalised Astana windows.
- **Weather forecasting (TimesBlock):** RMSE =  $0.034 \pm 0.002$  on HF weather.
- **Graph traffic (STGCN):** RMSE =  $0.479 \pm 0.002$  on HF traffic subgraphs.

- **Traffic density (PatchTST):**  $\text{RMSE} = 0.561 \pm 0.004$  on Astana.
- **Imputation (SAITS):** masked  $\text{MSE} = 0.640 \pm 0.009$ .
- **Energy (DLinear):**  $\text{RMSE} = 0.153 \pm 0.011$  on HF ETT.
- **Speed (iTransformer):**  $\text{RMSE} \approx 0.999 \pm 0.001$  — under the current training budget this slot does not meet deployment quality; it remains registered for routing experiments only.

Figure 5.2 shows seed dispersion; Figure 5.1 ranks forecasting RMSE. Together these provide multi-task evidence for **RQ3**.

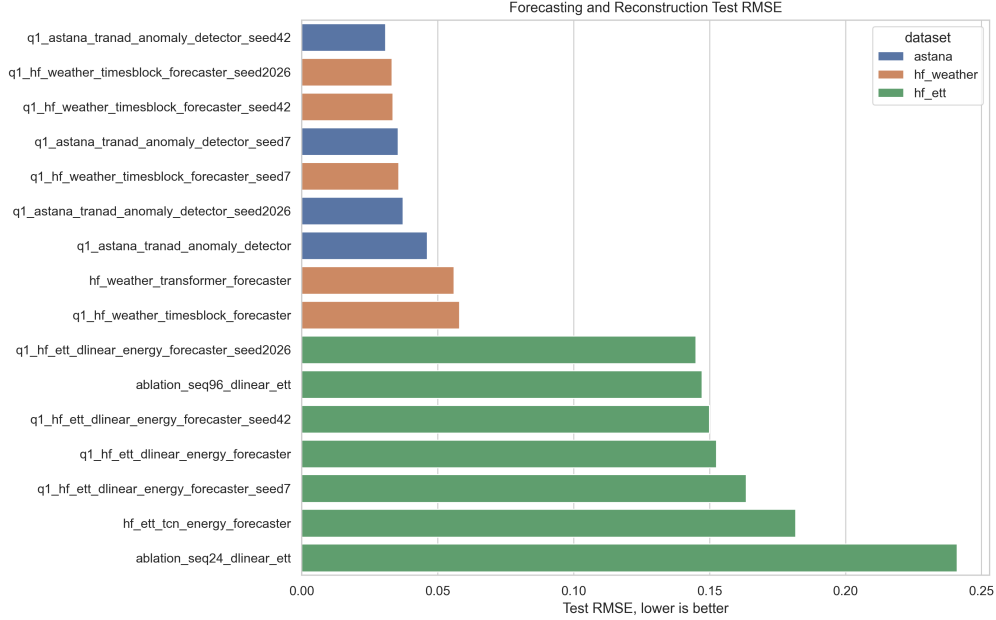


Figure 5.1: Forecasting and reconstruction leaderboard (test RMSE).

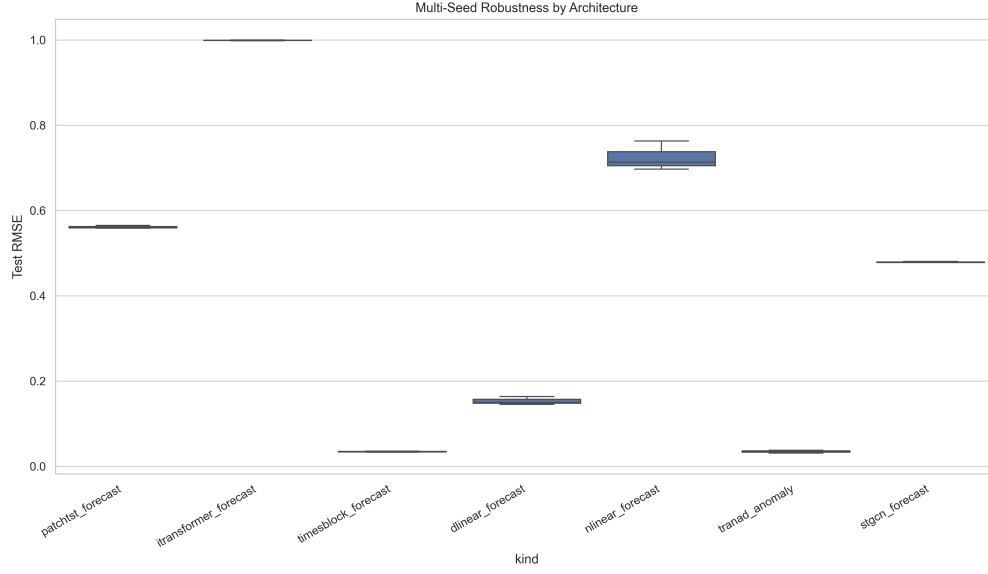


Figure 5.2: Multi-seed test RMSE dispersion for production portfolio models.

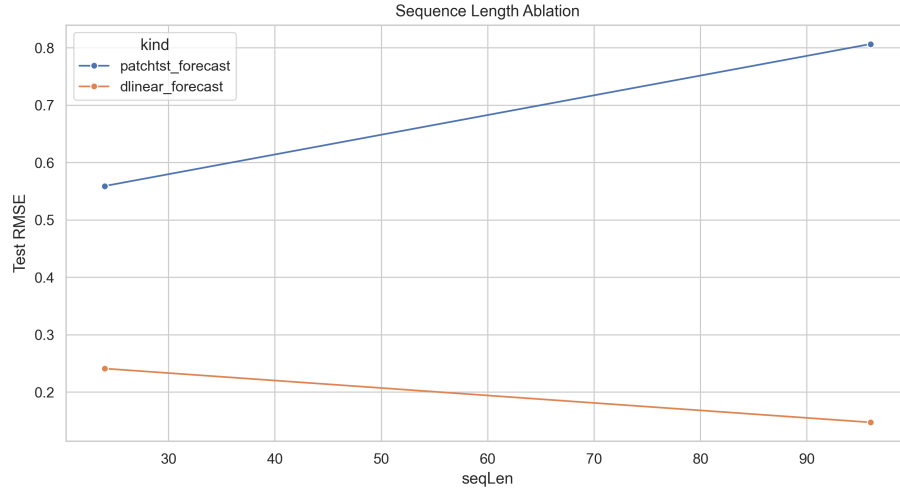


Figure 5.3: Sequence-length ablation ( $L = 24$  vs  $L = 96$ ).

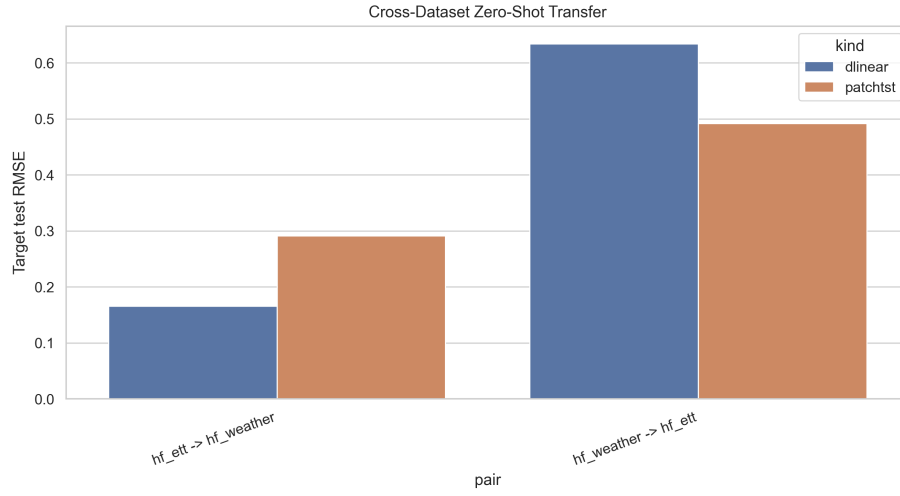


Figure 5.4: Cross-dataset transfer (train source  $\rightarrow$  test target).

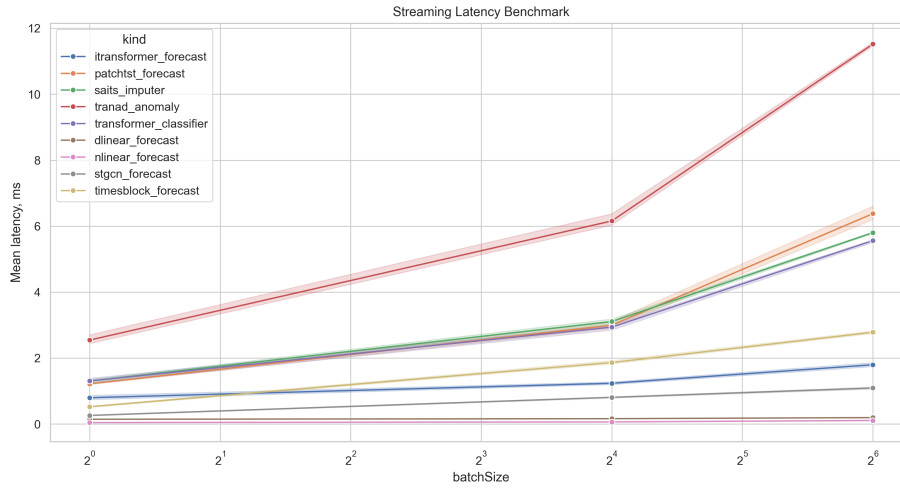


Figure 5.5: Streaming inference latency vs batch size (TorchScript, GPU).

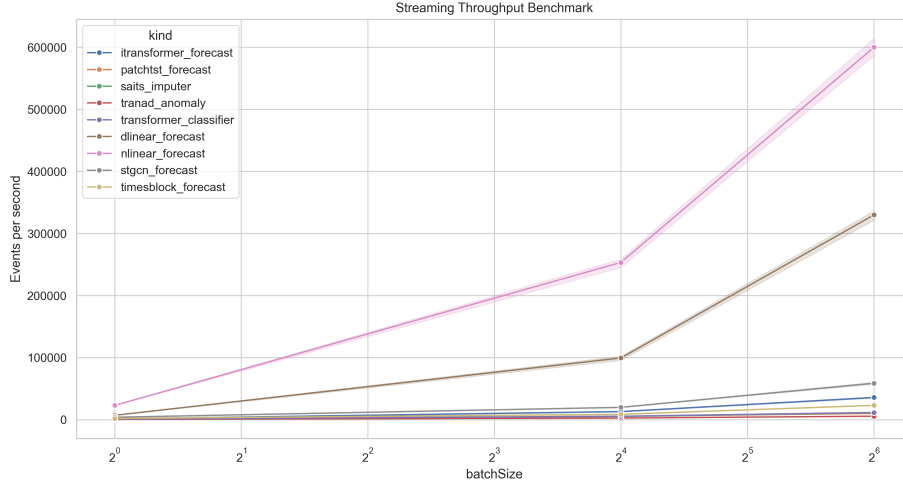


Figure 5.6: Streaming throughput (events/s) for production checkpoints.

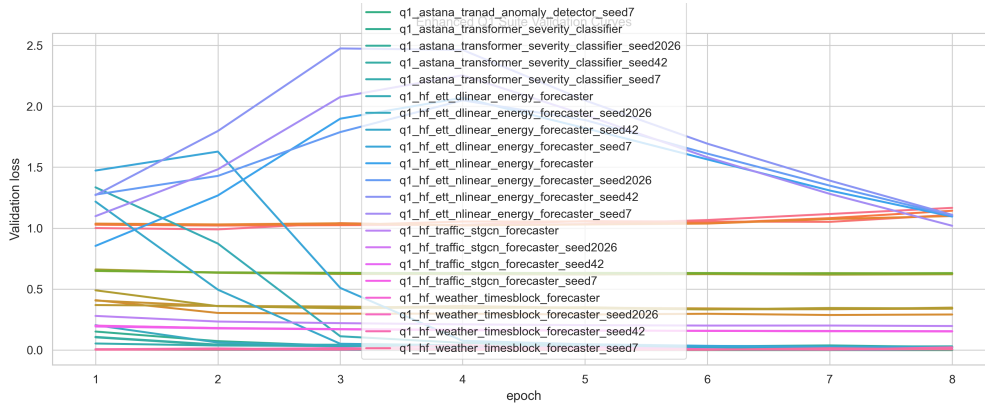


Figure 5.7: Validation loss curves during neural model training (representative runs).

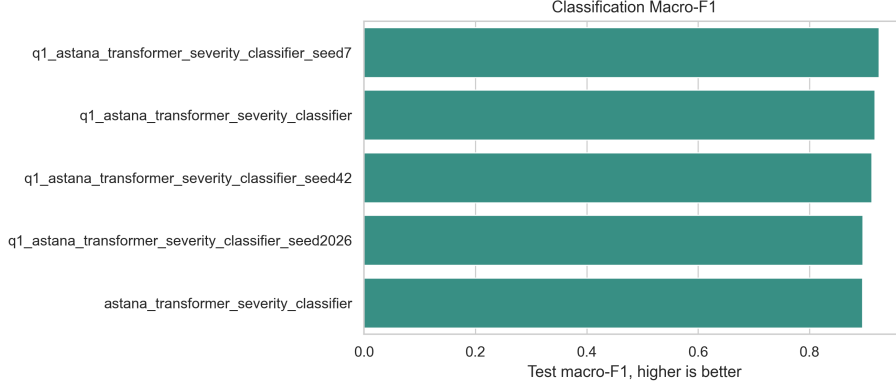


Figure 5.8: Severity classifier macro-F1 compared with classical baselines.

## 5.6 Ablations and Generalisation

### 5.6.1 Sequence length

Table 5.5 and Figure 5.3 compare  $L \in \{24, 96\}$ . Longer context helps DLinear on ETT (0.147 vs. 0.241 RMSE) but hurts PatchTST on Astana (0.807 vs. 0.559). Production default is  $L = 48$ .

### 5.6.2 Cross-dataset transfer

Table 5.6 and Figure 5.4 train on one HF bundle and test on another without fine-tuning. RMSE degrades versus in-domain runs, indicating domain shift between energy and weather telemetry.

### 5.6.3 Streaming inference latency

Figures 5.5 and 5.6 benchmark TorchScript exports. Mean single-event latency stays below 2 ms for DLinear and TimesBlock and below 2 s for large Transformers, informing smart-city deployment budgets.

## 5.7 Discussion

**Reproducibility.** Platform upload metrics and all neural aggregates in Table 5.4 trace to repository scripts and published HF weights.

**External validity.** Astana remains semi-synthetic; METR-LA/PEMS hold-outs are planned. HF cross-transfer is a first generalisation probe.

**Anomaly operationalisation.** TranAD reports reconstruction RMSE on normalised windows; event-level AUROC against injected anomalies is future work.

**Explainability.** Insight Engine metrics and optional LLM summaries complement reconstruction residuals and imputation masks; LLM fidelity is not formally evaluated.

**Routing.** Manual/Auto routing is **implemented** in the workbench (**RQ4**); match-rate and oracle comparisons are future work.

## 5.8 Summary

Four conclusions follow: (1) batch preparation achieves full quality scoring with sub-second processing on 30 000 records; (2) the Dockerised platform and Astana demo provide reproducible operational evidence; (3) seven of eight neural slots show usable held-out test performance under a shared protocol, with iTransformer flagged as weak; (4) registry routing **implements** modality-aware inference, pending quantitative evaluation.

# Chapter 6

## Conclusion and Future Directions

### 6.1 Summary of Contributions

This dissertation presents **Flowmatic**, an intelligent preparation and inference platform for urban transportation data:

- six-dimensional DQI and automated batch cleaning with measured latency;
- Dockerised batch and smart-city pipelines with WebSocket event delivery;
- eight Hugging Face neural checkpoints under a shared temporal hold-out protocol;
- Manual/Auto routing over a documented production registry;
- open reproduction scripts (`run_phase2_prep.py`, `run_phase3_core.py`).

### 6.2 Key Findings

Batch preparation on 30 000 Astana records completes in 406 ms with quality score 100/100. The Transformer severity model reaches  $\text{macro-F1} = 0.911 \pm 0.017$ ; TranAD anomaly reconstruction RMSE is  $0.035 \pm 0.004$  on Astana test windows. Compact forecasters (TimesBlock, DLinear) pair with low streaming latency; iTransformer speed forecasting requires retraining or registry retirement.

### 6.3 Limitations

- Semi-synthetic Astana data; no live agency deployment.
- METR-LA and PEMS-BAY ingested but not yet used as external hold-outs.
- Router match-rate and probabilistic forecast metrics (CRPS) not reported.
- TranAD evaluated on reconstruction RMSE, not event AUROC.
- LLM copilot summaries lack formal calibration.

### 6.4 Future Work

Priorities include METR-LA/PEMS hold-out rows, auto-router confusion analysis, AUROC for injected anomalies, extended training for weak slots, and operator studies on Insight Engine outputs. Federated coordination remains a demonstration path.

### 6.5 Concluding Remarks

Trustworthy ITS analytics requires auditable preparation connected to deployed models. Flowmatic demonstrates that quality assessment, multimodal neural inference, and smart-city orchestration can be delivered as one reproducible platform with published checkpoints and measured operational behaviour.

# Appendix A

## Author's Prior Publications

The present dissertation consolidates and extends the author's earlier peer-reviewed and presented work. The principal publications are cited in APA style below and referenced throughout the thesis.

1. Begisbayev, D., Sakhipov, A., Seiitbek, R., Mansurova, A., & Mansurova, A. (2024). Investigation of hybrid models and architectures for real-time adaptive passenger flow prediction. In *2024 7th International Conference on Data Science and Information Technology (DSIT)*, pp. 1–5. IEEE. <https://doi.org/10.1109/DSIT61374.2024.10880981>
2. Yedilkhan, D., Sakhipov, A., Seitbek, R., Mektepayeva, A., & Begisbayev, D. (2025). Intelligent assistant for data preparation automation in urban transportation management systems. *KazATC Bulletin*, 137(2), 310–324. <https://doi.org/10.52167/1609-1817-2025-137-2-310-324>
3. Mektepayeva, A., Begisbayev, D., Seiitbek, R., Khaimuldin, N., Sakhipov, A., & Rakhimzhanov, D. (2025). Adaptive machine learning algorithms for data processing in transportation systems. In *2025 IEEE 5th International Conference on Smart Information Systems and Technologies (SIST)*, pp. 1–8. IEEE. <https://doi.org/10.1109/SIST61657.2025.11139286>
4. Bakytuly, R. (2025). Real-time passenger flow prediction using automated data preparation. Paper presented at the 10th International Conference on Digital Technologies in Education, Science and Industry (DTESI), 2025.

5. Sakhipov, A., & Seiitbek, R. (2026). Event-driven microservices for incident detection and response in intelligent traffic systems. *International Journal of Information and Communication Technologies*, 7(1), 218–230. <https://doi.org/10.54309/IJICT.2026.25.1.014>

The thesis author (Ramazan Seiitbek) is listed as *Seiitbek, R.* on the IEEE and IJICT items above and as *Bakytuly, R.* on the DTESI 2025 entry, matching each publisher’s author metadata.

A companion manuscript on architectural patterns for Astana smart-traffic incident management (2026, *in preparation*) documents the semi-synthetic data generator used in the Astana case study; it is cited as unpublished work [?] and is not listed above because it is not yet published.

# Appendix B

## Reproducibility and Published Artifacts

### B.1 Repository Commands

```
# Phase 2 - datasets + production portfolio bundles
python models/paper/run_phase2_prep.py

# Phase 3 - multi-seed tables + streaming + HF upload
python models/paper/run_phase3_core.py

# Ablations and cross-dataset transfer
python models/paper_experiments.py
```

### B.2 Published Checkpoints

Eight models are listed in `models/reports/huggingface_model_manifest.json` and Table 4.2. Metrics aggregate in `models/paper/tables/multi_seed_aggregate.md`. Figures under `thesis/figures/results/` are regenerated via `scripts/render-thesis-figures.ps1`.

### B.3 Model Objectives

Forecast, anomaly, imputation, and classification losses are defined in Section 3.7. Local checkpoints reside under `models/checkpoints/`; Hub URLs use the `pushthetempo/flowmatic-`

namespace.

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